

Face HotPot: An Interactive Projection Device for Hot Pot

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ABSTRACT

Hot pot is a popular cuisine in China. It is also one of the first meal choices that young people elect to make together. However, during the preparation period, especially with new friends, there will always be a few moments of awkward silence while waiting for it to cook. To relieve this, we designed and developed Face HotPot, an interactive device for hot pots, consisting of a personalized dipping sauce matching system and interactive game. While waiting for the meal, users can get a personalized hot pot dipping sauce based on their facial features and get food by participating in the interactive projection game. The whole process of Face HotPot can continuously provide users with new chat topics and promote mutual understanding of each other's personalities, thus optimizing their eating and social experience with hot pot. In this paper, we demonstrate Face HotPot design motivation and system implementation.

CCS CONCEPTS

• Human-centered computing → Interaction devices.

KEYWORDS

Human-Food Interaction (HFI), Hot pot, Interactive game, Interactive device, Social behavior of young people.

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1 INTRODUCTION

Hot pot is one of the most popular dishes in China, especially in Sichuan and Chongqing. Hot pot is often the first choice for dinner when young people meet new friends [5]. However, for people having hot pot together for the first time, an embarrassing awkward silence often occurs while they wait. With our interactive hot pot system, we hope to alleviate the embarrassment by inviting users

to participate. For research, we first conducted a preliminary user study on young people in the Sichuan area, inquiring about their hot pot dining experiences via questionnaire and interview. Based on the results, we proposed the following design guidelines: (1) Deepen mutual understanding (2) Increase topicality. These guidelines directed our design of the project. We explored social interactions during hot pot dining experiences and used facial recognition technology to achieve personalized matching of tastes. At the same time, a fun desktop projection game using chopsticks to collect food can provide users with real-time new chat topics. Our interactive hot pot experience device, Face HotPot, is composed of a personalized dipping sauce matching system and an interactive projection game. In the process of experiencing this device and waiting for the food to cook, the user can gain social motivation. This demo explains the process of the preliminary user study and then describes the system design and implementation of Face HotPot.

2 RELATED WORK

2.1 Human-Food Interaction

The act of eating is, aside from its physiological function, an important social experience for humans and also an important influence on starting and maintaining interpersonal relationships [4]. Food can be used as an important medium for the transfer of social information, triggering emotional reactions, and the improvement of social connections [6]. Creative food interaction design can play a positive role in the health and dining experience of diners. Florian Echtler has used desktop projection to track dishes and calculate food consumption and influence the appetites of diners by simply adjusting the color of ambient lights [3]. The 'Mamagoto', designed by Takahiro Arakawa et al., has expanded the sensory experience of eating and helped children to enlarge their food knowledge [1]. Inspired by these prior researches, we focused on the regional food culture of hot pot to explore how interactive technology can help users improve their diet and social experience.

2.2 Interaction design with hot pot

There are a few research studies on interaction designs with hot pot. The hot pot restaurant 'Hai Di Lao' has carried out in-depth research in optimizing the effect of waiting time on customers. It gives customers a better experience if they wait alongside distributed fruits, snacks and drinks, nail salons, and other services [8]. Izumi Yagi et al. designed an interactive hot pot, Yaminabe YAMMY, that shares memories and feelings and builds strong relationships by explaining the associations with emails and photographs to different spices to adjust the hot pot taste [7]. Our study focuses on

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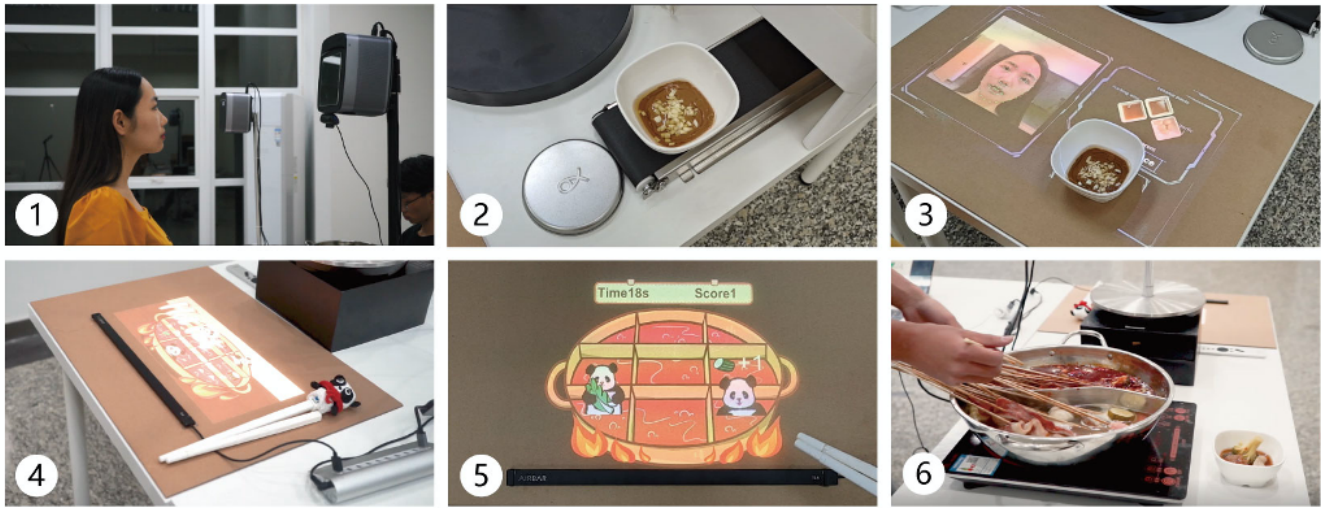


Figure 1: Interactive process: 1) look at the mirror, 2) get dipping sauce, 3) dipping sauce display interface, 4) play ‘Catch Panda’, 5) pick up virtual food, and 6) get edible hot pot food.

the design of the interactive experience during the waiting time for hot pots. We want to explore the possibilities of using design interactions to improve the dining experience and enhance users’ social interaction.

3 PRELIMINARY USER STUDY

We conducted an online questionnaire survey and several interviews with young people aged 20–25, to explore their dietary experiences with hot pot. The survey lasted two weeks and we collected 112 questionnaires and 12 interviews. We distributed and collected questionnaires on a questionnaire platform and invited target users to fill out via social media and local bulletin board systems. By analyzing quantitative data, we learned about the participants’ hot pot preferences, the purpose of having hot pot with friends, and the reasons why they felt embarrassed while awaiting its preparation. Based on the statistical overview of the questionnaire platform, our researchers conducted a cross-data analysis of related questions.

After the questionnaire survey, we conducted semi-structured interviews averaging 20 minutes each, with 12 participants, of whom five were women. The purpose of the interviews was to explore the embarrassing experiences of participants while preparing hot pot. We started the interviews with the story of ‘An experience of eating hot pot with new friends’ and discussed the challenges and social interactions they faced while waiting. All interviews were recorded and transcribed with written and oral consent. Our work was guided by Anselm Straus’s [2] quantitative research methods to organize and encode interview manuscripts. Based on the above studies, we considered the following design goals:

Features of regional cuisine. There are several indispensable parts of hot pot culture, such as the dipping sauce and chopsticks. People’s choices of dipping sauce with hot pot showed obvious differences, so we matched suitable dipping sauces based on the personality

attributes of different people. We chose panda as the cartoon character in the game and designed and developed chopsticks with sensors as the input tool.

Deepen mutual understanding. A personalized dipping sauce matching system was used to display different participant attributes and match different flavors of hot pot dipping sauces based on these attributes. Visualizing the personality characteristics and taste preferences of the participants helped to create topics for them to understand each other better.

Increase topicality. Games are one of the most popular interactive methods for users. Playing the same game can promote communication among young people. Therefore, we designed an interactive desktop game—‘Catch the Panda’. In the game, users controlled chopsticks to interact with the panda on the interface. Game scores can be exchanged for edible hot pot food. The game process provided new chat topics for discussion.

4 DESIGN OF FACE HOTPOT

4.1 System Overview

We designed Face HotPot, an interactive device for the hot pot to optimize the waiting and social experience. It consists of a personalized dipping sauces matching system and interactive game. The dipping sauce matching system determines sauce flavors through facial detection, which matches dipping sauce to physiognomic facial features. In the interactive game, participants can use chopsticks to pick up the cartoon panda projected on the table and get the corresponding food. Through the interaction of the two parts, Face HotPot can promote players’ mutual understanding, provide them with new chat topics, and further optimize their experience of dining and socializing.

4.2 Demonstration

The interactive process of Face HotPot works as follows(Fig 1): Participants are asked to stand in front of a mirror and wait for the camera to collect their facial information. The computer takes facial data and analyses it. After the analysis, the system matches the physiognomic features with the dipping sauce, and the most highly matched dipping sauce is then handed to the participant on a conveyor belt. Participants can later place the dipping sauce in the recognition area, the ratio of ingredients and meanings of the dipping sauce will be projected to the desktop by the system. On the other side of the device, participants can use the chopsticks to play the interactive game of catching the panda and also the hot pot food that matches the game score.

Personalized dipping sauces matching system: Based on the traditional Chinese Physiognomy culture and nine-type personality theory, we analyzed the users' personality attributes and matched them with corresponding hotpot dipping sauces. For example, people with larger mouths tend to be more cheerful, while extroverts are more likely to like sweetness. The system will first extract the features from the facial image, then map these features to the closest related personality. After finding the personality, the system will automatically select a dipping sauce based on the association between the personality and its taste preferences and hand it to the user. We select six common hotpot dipping sauces (Fig 2): traditional sauce, classic hemp sauces, garlic oyster sauce, chili oil and garlic sauce, satay sauce, and dried peppers. These dipping sauces are composed of 18 different seasonings, including sesame sauce, hot sauce, onion, ginger, and garlic.



Figure 2: Six kinds of dipping sauce:1) traditional sauce, 2) classic hemp sauce, 3) garlic oyster sauce, 4) chili oil and garlic sauce, 5) satay sauce, 6) dried peppers

Interactive projection game - 'Catch Panda': In 'Catch Panda', the projector will display a nine-block hot pot on a piece of Kraft paper, in which pandas randomly appear holding different hotpot foods. Participants can use chopsticks to pick up the pandas in 20 seconds. At the end of the game, participants can exchange the virtual food obtained in the game for real food, such as ham, mushrooms, lotus root, and so on. However, if participants pick up the panda that holds the hot red pepper and numbing Sichuan pepper, the game will end immediately, and the seasoning ingredients such as ginger and garlic will not be counted. This restores the dining scene of

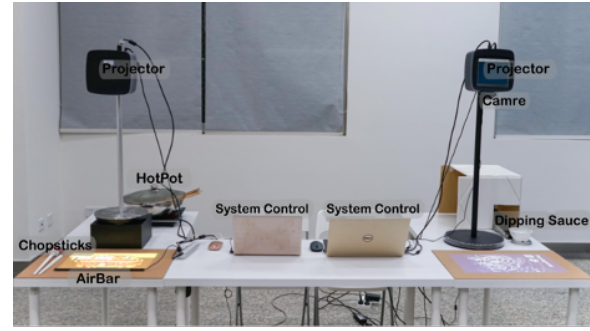


Figure 3: Prototype of Face HotPot

hotpot and enables players to get quick bites during the waiting time.

4.3 System Implementation:

The Face HotPot prototype consists of a system that matches and delivers dipping sauce and an interactive game of catching the panda (Fig 3). To capture the facial traits from the participants, we used the Deep Alignment Network algorithm to detect key face points and match the corresponding dipping sauce with the physiognomy scale. The conveyor belt on the table conveys the dipping sauce to the participants. In the interactive game, chopsticks are used to detect food picking, which is achieved by the magnetron module and magnetic beads placed separately on chopstick heads. It can accurately detect the mutual contact of the chopstick heads. The position of the chopstick is accurately detected with the Airbar Touchscreen sensor on the Kraft paper.

5 CONCLUSION

In this demo, we describe the design motivation and system development of Face HotPot. We first collected the needs of young people to choose the hot pot for gatherings through questionnaires and interviews and proposed design guidelines to steer us in designing Face HotPot. Then, we explained the interactive process of the system. We hope that this research could optimize the eating and social experience of young people while enjoying the hot pot. However, the project still has some limitations. It has only been conducted as prototype testing in the laboratory without having run a full user study, due to Covid-19. In the future, we plan to set up our device in a hot pot restaurant and invite customers to experience our system to test and verify our design goals. We hope that this research can contribute to user experience in the field of human-food interaction. Also, we will continue to improve the system design and further explore more interesting hot pot interactions.

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